Weiyu Li

Harvard University
School of Engineering and Applied Science
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Research interests

High-dimensional statistics and machine learning theory, random matrix theory, distributed optimization

Education

Ph.D., Applied Mathematics, Harvard University 09/2020–05/2027 (expected)

· Advisor: Yue M. Lu Committee: Alan Edelman, Jun S. Liu, Lucas B. Janson

B.Sc., University of Science and Technology of China (USTC) 09/2016–07/2020

· School of Gifted Young

· Major: Mathematics and Applied Mathematics (Probability and Statistics Track)

· GPA: 4.11/4.30 (94.14/100.00) Major GPA: 4.25/4.30 (96.50/100.00)

Research Experience

Research Assistant, Department of Statistics and Finances, USTC 09/2020–08/2021

· Advisor: Canhong Wen

Research Intern, Harvard University 06/2019–09/2019

· Advisor: Yue M. Lu

Publications

Journals

1. Jie Peng, **Weiyu Li**, Stefan Vlaski, Qing Ling. Topology-Independent Learning Error of the Weighted Mean Under Label Poisoning Attacks in Heterogeneous Decentralized Learning. *IEEE Transactions on Signal and Information Processing over Networks (T-SIPN)*, 2026.
2. Jie Peng, **Weiyu Li**, Stefan Vlaski, Qing Ling. Mean Aggregator is More Robust than Robust Aggregators under Label Poisoning Attacks on Distributed Heterogeneous Data. *Journal of Machine Learning Research (JMLR)*, 26(27):1–51, 2025.
3. Runyu Zhang, **Weiyu Li**, and Na Li. Optimal Network Control of Spatially Exponential Decaying Linear Quadratic Regulator. *Automatica*, 2025.
4. Jie Peng, **Weiyu Li**, Qing Ling. Byzantine-robust Decentralized Stochastic Optimization with Stochastic Gradient Noise-independent Learning Error. *Signal Processing*, 2024.
5. Feng Lin, **Weiyu Li**, Qing Ling. Stochastic Alternating Direction Method of Multipliers for Byzantine-robust Distributed Learning. *Signal Processing*, 2022.
6. **Weiyu Li**, Zhaoxian Wu, Tianyi Chen, Liping Li, Qing Ling. Communication-censored Distributed Stochastic Gradient Descent. *IEEE Transactions on Neural Networks and Learning Systems (T-NNLS)*, 33(11):6831–6843, 2022.

7. Jie Peng, **Weiyu Li**, Qing Ling. Byzantine-Robust Decentralized Stochastic Optimization over Static and Time-varying Networks. *Signal Processing*, 2021.

8. **Weiyu Li**, Yaohua Liu, Zhi Tian, Qing Ling. Communication-censored Linearized ADMM for Decentralized Consensus Optimization. *IEEE Transactions on Signal and Information Processing over Networks (T-SIPN)*, 2020.

Conferences

1. Xiaomin Li, Mingye Gao, Zhiwei Zhang, Jingxuan Fan,**Weiyu Li**. Data-adaptive Safety Rules for Training Reward Models. *International Conference on Learning Representations (ICLR) Workshop on Bidirectional Human-AI Alignment*, 2025.

2. Jie Peng, **Weiyu Li**, Qing Ling. Mean Aggregator Is More Robust than Robust Aggregators under Label Poisoning Attacks. *International Joint Conference on Artificial Intelligence (IJCAI)*, 2024.

3. Runyu Zhang, **Weiyu Li**, and Na Li. On the Optimal Control of Network LQR with Spatially-exponential Decaying Structure. *American Control Conference (ACC)*, 2023.

4. Runyu Zhang, Yang Zheng, **Weiyu Li**, and Na Li. On the Relationship of Optimal State Feedback and Disturbance Response Controllers. *World Congress of the International Federation of Automatic Control (IFAC)*, 2023.

5. Jie Peng, **Weiyu Li**, Qing Ling. Variance Reduction-boosted Byzantine Robustness in Decentralized Stochastic Optimization. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2022.

6. Jie Peng, **Weiyu Li**, Qing Ling. Variance Reduction-Boosted Byzantine Robustness in Decentralized Stochastic Optimization. *Chinese Control Conference (CCC)*, 2022.

7. Jie Peng, **Weiyu Li**, Qing Ling. Byzantine-Resilient Decentralized Stochastic Optimization over Static and Time-Varying Networks. *Chinese Control Conference (CCC)*, 2021.

8. Feng Lin, Qing Ling, **Weiyu Li** and Zhiwei Xiong. Stochastic ADMM for Byzantine-robust Distributed Learning. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2020.

9. **Weiyu Li**, Yaohua Liu, Zhi Tian, Qing Ling. COLA: Communication-censored Linearized ADMM for Decentralized Consensus Optimization. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2019.

Preprints

1. Canhong Wen, Ruipeng Dong, Xueqin Wang, **Weiyu Li**, Heping Zhang. Simultaneous Best Subset Selection and Dimension Reduction via Primal-dual Iterations. *ArXiv: 2211.15889.*

Industry Internships

Quantitative Research Internship, Scientech Research LLC 06/2024–08/2024

· Futures market research Advisor: Rui Wei

Quantitative Research Internship, Citadel Securities 06/2023–08/2023

· Options market impact Advisor: Zongxi Li

Teaching Experiences

Teaching Fellow, AM 101: Statistical Inference for Scientists and Engineers 2026 Fall
Teaching Fellow, AM/ES 254: Math. of High-Dimensional Information Processing & Learning 2022, 2024, 2025 Fall
Teaching Fellow (**Excellent Teaching Award**), ES 150: Probability with Engineering Appl. 2022, 2024, 2025 Spring
Teaching Fellow, AM/ES 158: Introduction to Optimal Control and Estimation 2023 Fall
Teaching Assistant (**Top 10 Teaching Award**), Convex Optimization 2020 Fall
Teaching Assistant (Undergraduate), Nonparametric Statistics (Graduate Course) 2019 Fall
Teaching Assistant (Undergraduate), Real Analysis 2019 Spring
Teaching Assistant (Undergraduate), Mathematical Statistics 2018 Fall

Educational Experiences

Stat. & Trustworthy AI for Cross-Domain Acceleration (STAI-X), Harvard University 07/2026–08/2026
Pre-Concentration Advisor, Harvard University 09/2024–05/2026
Princeton Machine Learning Theory Summer School, Princeton 08/2024
Summer School on Statistical Physics of Machine learning, Les Houches School of Physics 07/2022
Master Classes in Probability, IRMA, University of Strasbourg 01/2020

Awards and Honors

Excellent Teaching Award, Harvard University 06/2022
Outstanding Undergraduate Thesis, USTC 06/2020
Outstanding Undergraduate Award, USTC 05/2020
The 39th Guo Moruo Scholarship (**top scholarship** in USTC) 10/2019
Team Bronze Medal (**top 2** in Prob Stat), *10th S.-T.Yau College Student Mathematics Contests* 05/2019
Tang Lixin Scholarship (**top** 0.3% students in USTC) 09/2018

Professional Services

Program Committee Member, AAAI Conference on Artificial Intelligence (AAAI) 2026, 2027
Course Monitor (CE_17), Joint Statistical Meetings (JSM), Boston, MA 2026
Reviewer for International Conference on Learning Representations (ICLR) 2026
Reviewer for International Conference on Machine Learning (ICML) 2025
Reviewer for International Conference on Artificial Intelligence and Statistics (AISTATS) 2025
Reviewer for Mathematics of Control, Signals, and Systems 2025
Reviewer for Conference on Neural Information Processing Systems (NeurIPS) 2022, 2024, 2025
Reviewer for IEEE Statistical Signal Processing (SSP) Workshop 2023
Reviewer for IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2022
Reviewer for Signal Processing, IEEE Signal Processing Letters (SPL) 2021

Languages and Skills

Computer Skills: Python, MATLAB, Julia, R
Language: Chinese (native), English (fluent)